VICTORY 2027

SOUND

LECTURE-04 PHYSICS

Shailendra sir



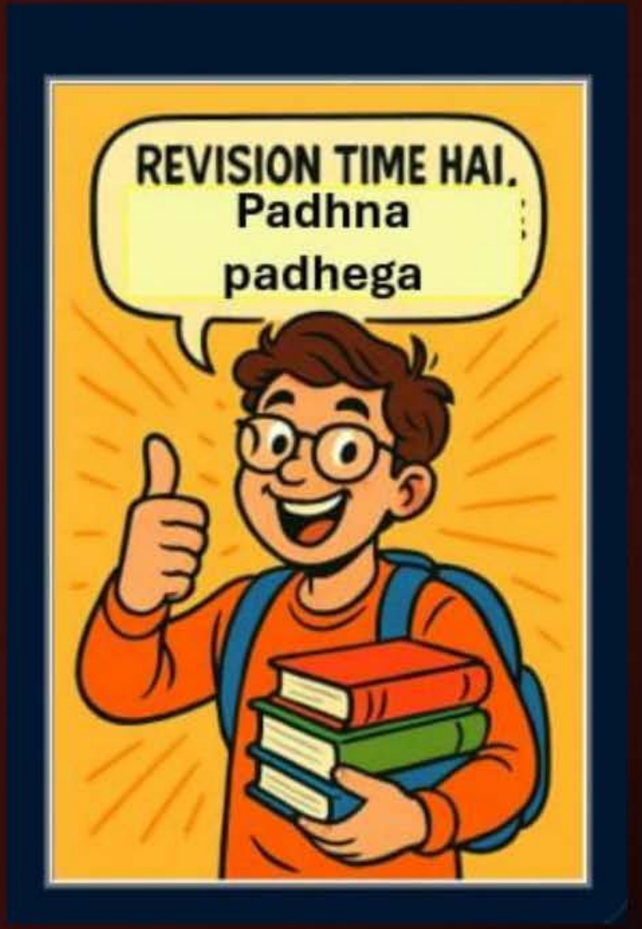
Topics To Be Covered

1. REVISION
2. CHARACTERISTICS OF SOUND
3. MUSIC & NOISE
4. QUESTIONS


SHAILENDRA SIR
JOIN MY OFFICIAL TELEGRAM CHANNEL
JOIN NOW FOR PDF NOTES
@PHYSICSBY_SHAILENDRA_SIR
Physics Wallah



Revision



CHARACTERISTICS OF SOUND

Two sounds can be distinguished from one another by the following three different characteristics:

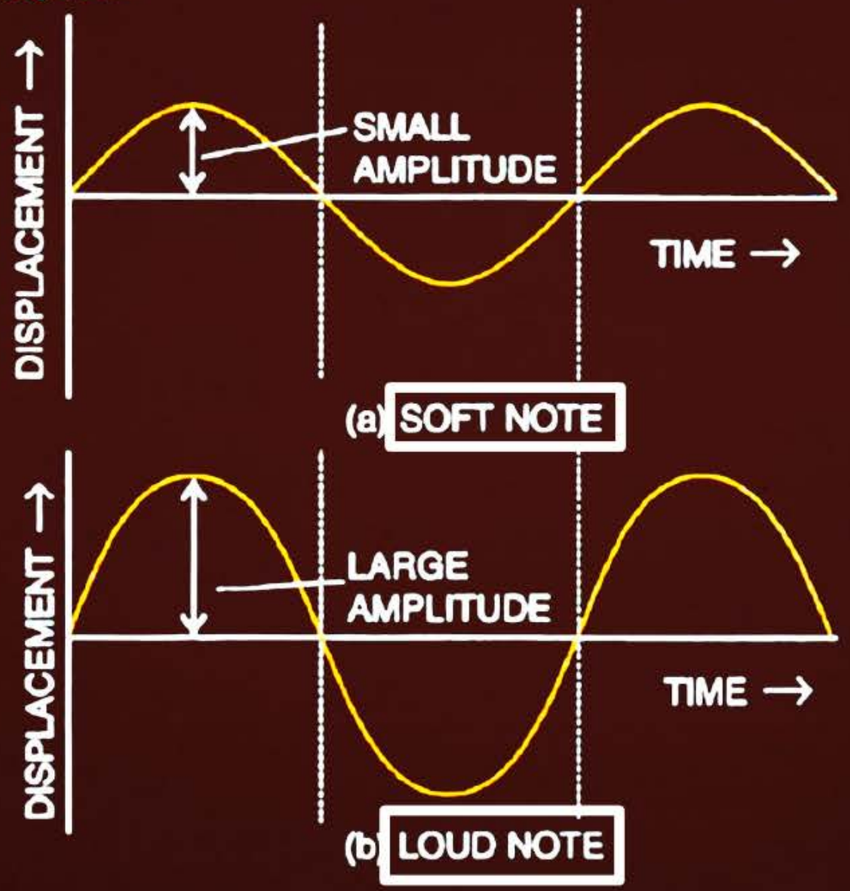
(1) Loudness → Amplitude
(2) Pitch (or shrillness) → Frequency
(3) Quality (or timbre).
↳ Waveform

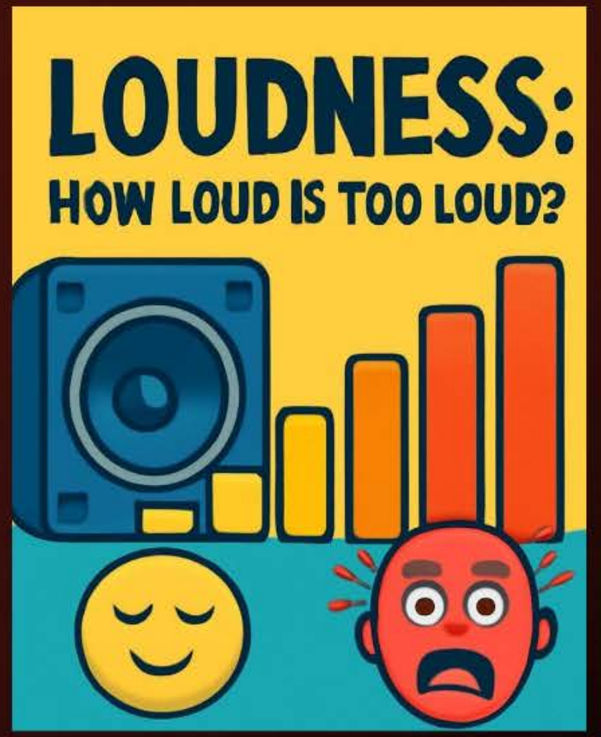


Loudness

Loudness is the characteristic by virtue of which a loud sound can be distinguished from a faint one, both having the same pitch and quality. Loudness of sound depends on the amplitude of the wave.

$$\text{Loudness} \propto A^2$$



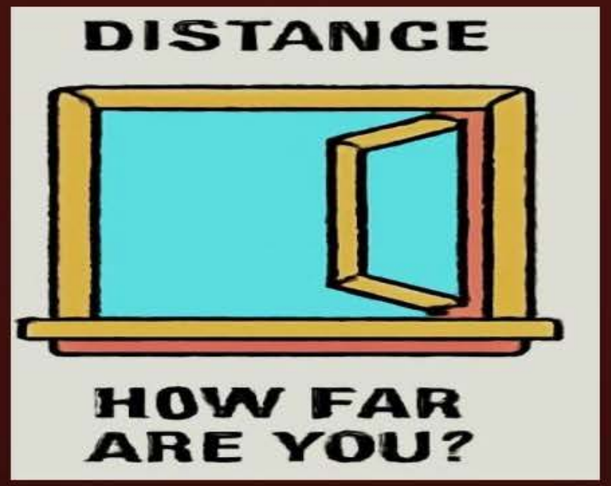


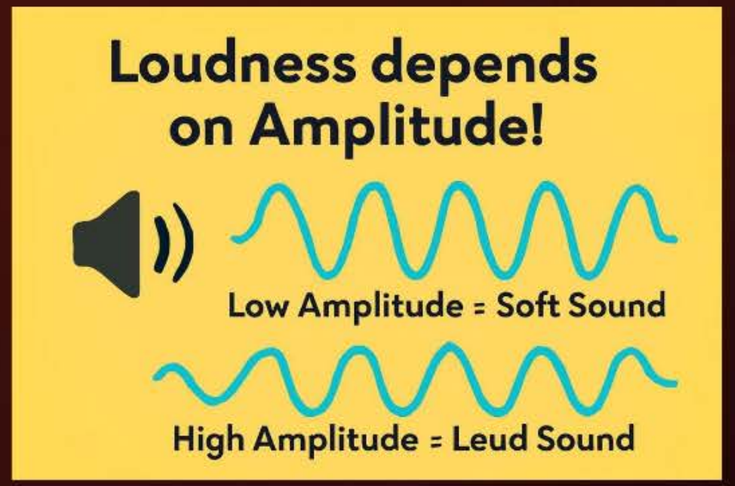
Factors Affecting the Loudness of Sound

(i) **Loudness is proportional to the square of the amplitude:** When a body vibrates with a greater amplitude, it sends forth a greater amount of energy and hence the energy received by the eardrum is large, so the sound appears louder. $L \propto A^2$

(ii) **Loudness varies inversely as the square of distance from the source:** If the listener is close to the source of sound, he hears it quite louder, but if he is far away, the sound becomes feeble. If he moves further away from the source, a stage may reach when the sound becomes inaudible. Thus, closer the source, louder is the sound.

$L \propto \frac{1}{d^2}$



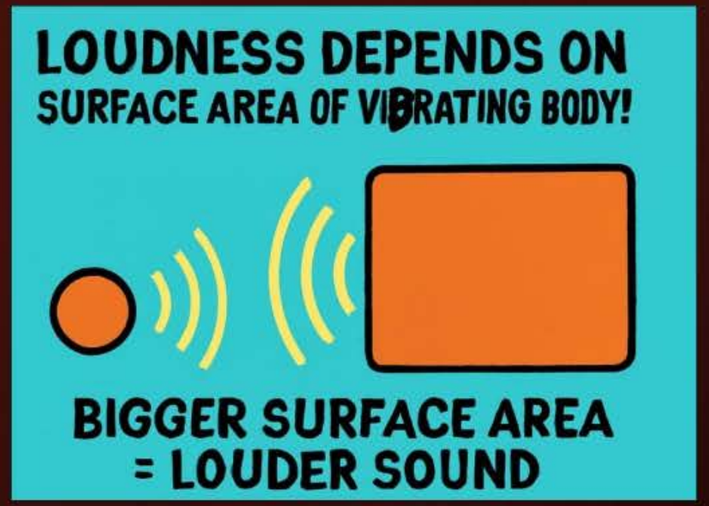


Factors Affecting the Loudness of Sound

(iii) **Loudness depends on the surface area of the vibrating body:** A large vibrating area sends forth a greater amount of energy. Hence larger the surface area of the vibrating body, louder is the sound heard.

- For this reason, the bell in schools, temples etc. is made big in size.

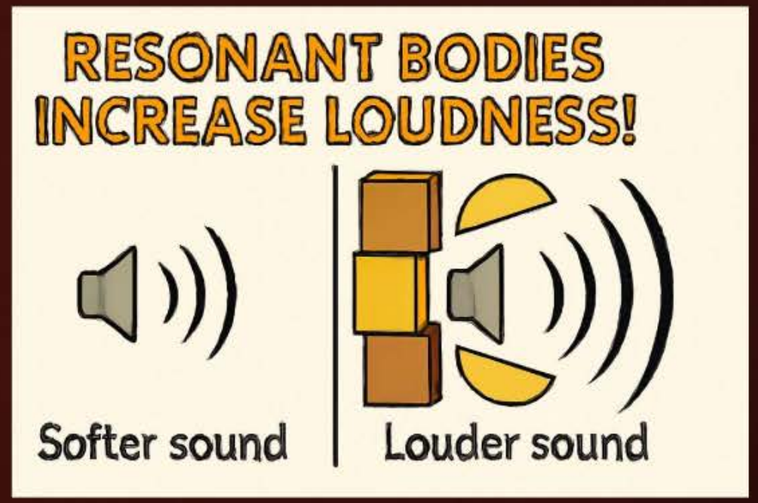
When a tuning fork is sounded in air, the sound given by it is feeble, but when it is placed on a table top, the sound becomes much louder. The reason is that the table top provides comparatively a large surface area of air to vibrate and so the sound energy reaching our ears increases. Similarly in case of the stringed instruments, a sound chamber is provided to increase the volume of air to vibrate and hence the loudness of sound increases.



Factors Affecting the Loudness of Sound

(iv) **Loudness depends on the density of the medium:** More the density of medium, more is the loudness.

(v) **Loudness depends on the presence of resonant bodies:** The presence of resonant bodies near the vibrating body increases the loudness of sound.



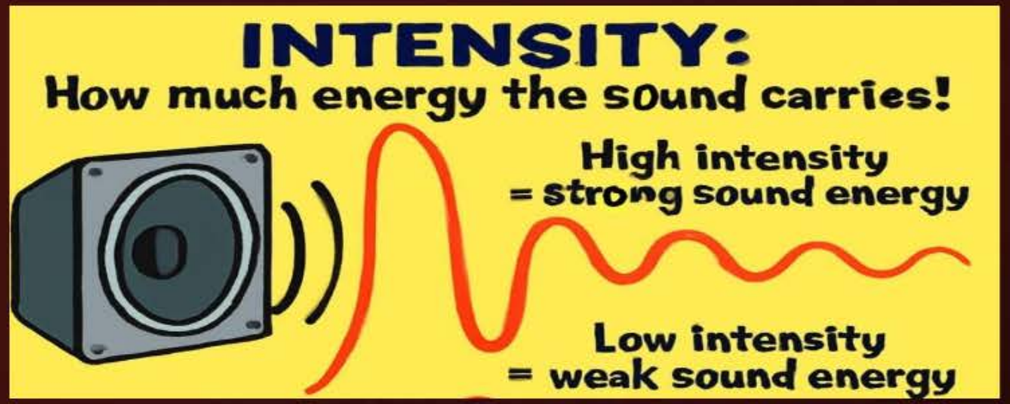


Intensity

$$I = \frac{E}{A \cdot t}$$

The intensity of a sound wave at a point of the medium is the **amount of sound energy passing per second normally through unit area** at that point. Its unit is **watt per metre²** ($\mathrm{Wm^{-2}}$).

- The intensity of ordinary sound is nearly $10^{-6}\mathrm{Wm^{-2}}$, while minimum intensity of sound audible to ears is $10^{-12}\mathrm{Wm^{-2}}$.
- Greater the energy carried by a sound wave, greater is the intensity of sound.
- The intensity of a sound wave in air is proportional to (a) the square of the amplitude of vibrations, (b) the square of the frequency of vibrations, and (c) the density of air.



Subjective Nature of Loudness and Objective Nature of Intensity

Depends on observer.

Measurable Quantity

The loudness of a sound depends on

(i) The intensity i.e., the energy conveyed by the sound wave near the eardrum of the listener,

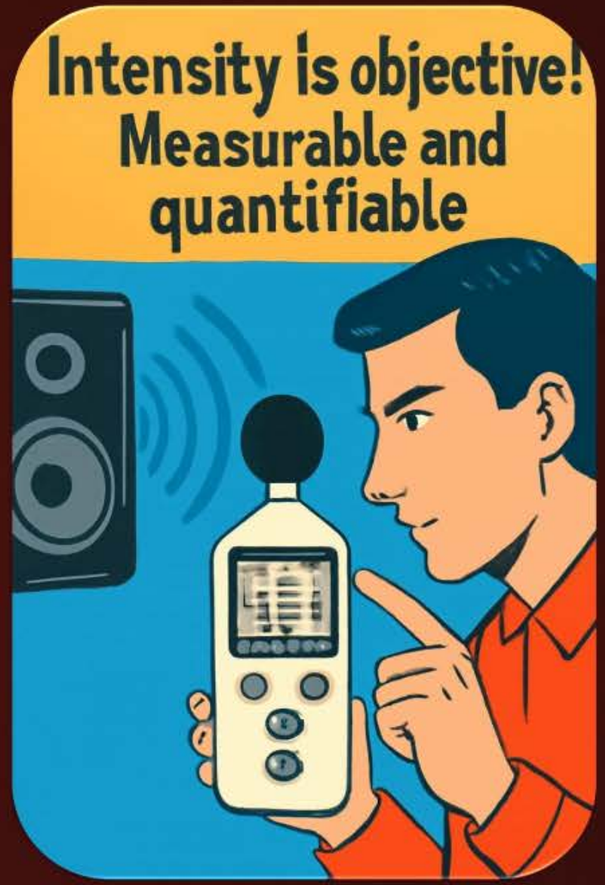
(ii) The sensitivity of the ears of the listener.

Thus, loudness of sound of a given intensity **may differ from listener to listener.**

- Further, two sounds of the same intensity, but of different frequencies may differ in loudness even to the same listener because the **sensitivity of ears is different for different frequencies**.
- **For normal ears, the sensitivity is maximum at the frequency 1 kHz.**

Thus, for a sound wave loudness has a subjective nature, while intensity, being a measurable quantity, has an objective nature.


LOUDNESS IS SUBJECTIVE!
DEPENDS ON LISTENER
NOT LOUD AT ALL
Too loud!


Intensity is objective!
Measurable and quantifiable

Relationship between Loudness and Intensity

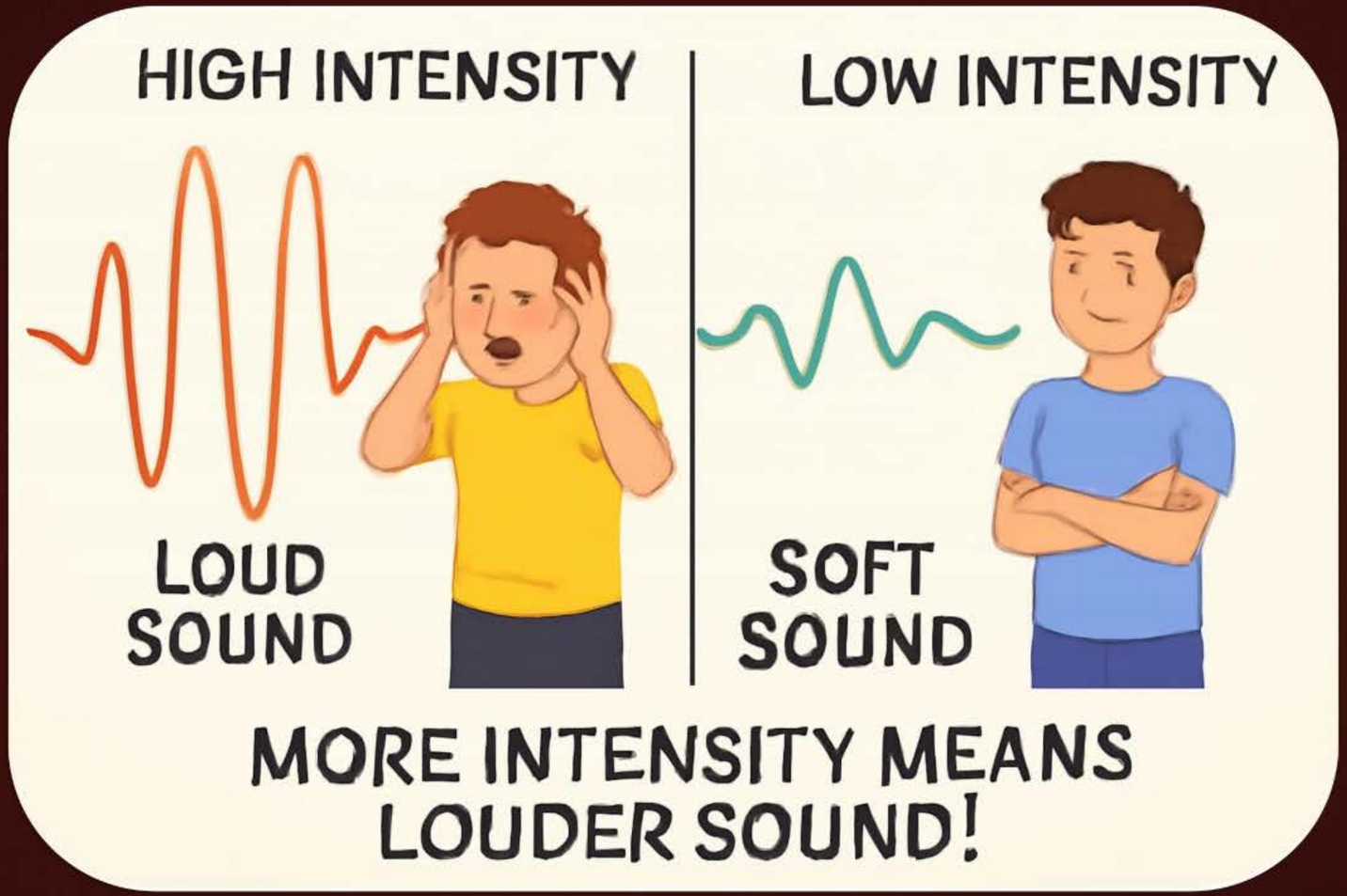
Although loudness of sound depends on its intensity, but loudness is not same as intensity. **Intensity is a measurable quantity, while loudness is a sensation**. Experimentally, Weber and Fechner established a relationship between the loudness L and intensity I which is given as :

$$L = K \log_{10} I$$

where K is a constant of proportionality. **Obviously, loudness increases with the increase in intensity, but not in the same proportion.**

Units of loudness and sound level (phon and decibel) :

- The unit of loudness is phon.
- The level of sound is expressed in decibel (dB).
- The loudness of a sound in phon is the loudness in decibel of an equally loud pure sound of frequency 1 kHz .
- Thus both the phon and the decibel scales agree at the frequency 1 kHz .


HIGH INTENSITY
LOUD SOUND
LOW INTENSITY
SOFT SOUND
MORE INTENSITY MEANS LOUDER SOUND!



Relationship between Loudness and Intensity

If at a given frequency, I_1 and I_0 are the intensities of two sounds of which loudness are L_1 and L_0 respectively, then

$$L_1 = K \log_{10} I_1$$
$$L_0 = K \log_{10} I_0$$

Hence difference in loudness of the two sounds is

$$L = L_1 - L_0 = K(\log_{10} I_1 - \log_{10} I_0)$$

$$\mathbf{L = K \log_{10} \frac{I_1}{I_0}}$$

If I_0 is the minimum intensity $(= 10^{-12} \mathrm{Wm}^{-2})$ of audible sound at frequency 1 kHz, then L is called the **sound level**.

Taking $K = 1$, the sound level is written

$$\boxed{\mathbf{L = \log_{10}\left(\frac{I_1}{I_0}\right)}} \text{ bel}$$

Relationship between Loudness and Intensity

$$L = \left\{10 \log_{10}\left(\frac{I_1}{I_0}\right)\right\} \times \frac{1}{10} \text{ bel}$$

If we write $\frac{1}{10}$ bel as decible (dB), then

$$\boxed{L = 10\log_{10}\left(\frac{I_1}{I_0}\right)} \text{ decibel}(dB)$$

Here bel or decibel (dB) is the unit of loudness level, named after the scientist **Alexander Graham Bel**, the inventor of telephone.

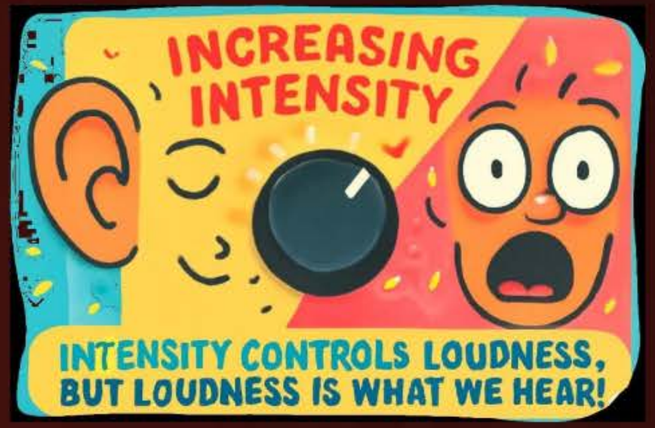
Relationship between Loudness and Intensity

Now if $L = 1\text{dB}$,

$$10\log_{10}\left(\frac{I_1}{I_0}\right) = 1$$

$$\log_{10}\left(\frac{I_1}{I_0}\right) = \frac{1}{10}$$

$$\frac{I_1}{I_0} = \text{Antilog}\left(\frac{1}{10}\right) = \text{Antilog}(0.1) = 1.26$$



Thus 1 dB is defined as the change in level of loudness when the intensity of sound changes by 26%.



$$\text{Intensity} = \frac{\text{Sound Energy}}{\text{Area} \cdot \text{Time}}$$

Loudness ↑ Intensity ↑

$$L = K \log_{10} I \quad \text{——} ①$$

For Min. Intensity of Sound,

$$L_0 = K \log_{10} I_0 \quad \text{——} ②$$

Change in loudness,

$$\Delta L = L - L_0 = K(\log I - \log I_0)$$

$$\Delta L = K \log \frac{I}{I_0}$$

$$\boxed{L = \log \frac{I}{I_0}} \text{ bel}$$

$$\boxed{L = 10 \log \frac{I}{I_0}} \text{ dB}$$

$$① = 10 \log \frac{I}{I_0}$$

$$\frac{1}{10} = \log \frac{I}{I_0}$$

$$\text{antilog}(0.1) = \frac{I}{I_0}$$

$$I = I_0 (\text{antilog } 0.1)$$

$$\boxed{I = 1.26\ I_0}$$

26 % increase.

1 dB change in loudness is equivalent to 26 % rise in intensity of sound.



Relationship between Loudness and Intensity

S.No.	Source of sound	Loudness (in dB)	Intensity (W/m^2)
1.	Threshold of hearing	0	10^{-12}
2.	Normal breathing	10	$10^{1} \times 10^{-12} = 10^{-11}$
3.	Soft whisper (at 5m)	30	$10^{3} \times 10^{-12} = 10^{-9}$
4.	Normal conversation	60	$10^{6} \times 10^{-12} = 10^{-6}$
5.	Busy traffic	70	$10^{7} \times 10^{-12} = 10^{-5}$
6.	Average factory	80	$10^{8} \times 10^{-12} = 10^{-4}$
7.	Threshold of pain	130	$10^{13} \times 10^{-12} = 10^{1}$



Noise Pollution

Disturbance produced in the environment due to undesirable loud and harsh sound, of level above 120 dB, from various sources such as loudspeaker, siren, moving vehicles, etc. is called noise pollution.

- A constant hearing of sound of level above 120 dB can cause headache and permanent damage to the ears of the listener.
- The safe limit of level of sound for hearing is from 0 to 80 dB.
- Sound of level 30 dB to 10 dB has a soothing sensation.
- While the sound of level 0 dB represents the lower limit of hearing. $(10^{-12}\ W/m^2)$

Note

The limit of hearing should not be confused with the limit of audibility. The limit of audibility refers to the frequency of sound 20 Hz to 20 kHz which a man can hear, while the limit of hearing is the lowest level of loudness (=0 dB) which a man can hear.

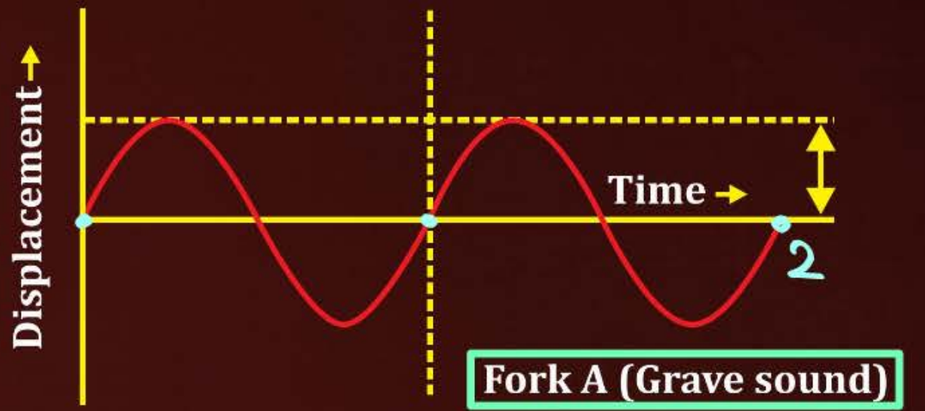

NOISE POLLUTION

Pitch (or Shrillness) and Frequency

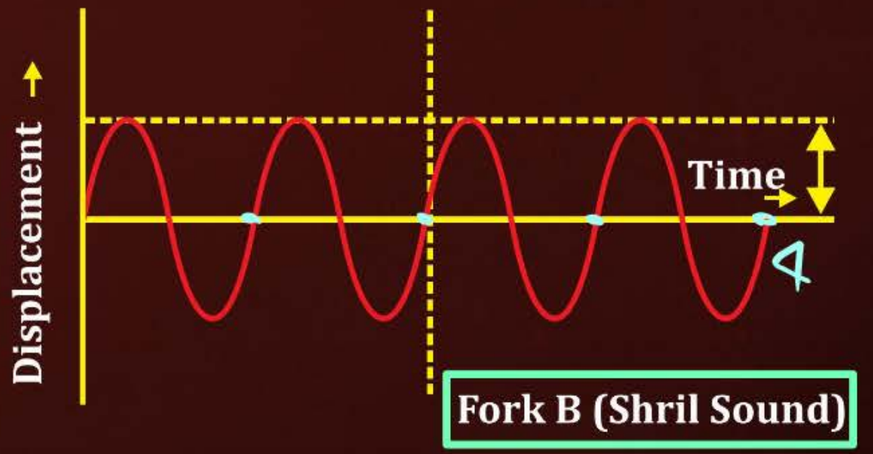
Pitch is that characteristic of sound by which an acute (or shrill) note can be distinguished from a grave or flat note.

- Pitch refers only to the musical sounds and each musical note has a definite pitch. If the pitch is high, the sound is shrill and if the pitch is low, the sound is flat.
- In a tape recorder, **bass and treble** refer to low and high pitch respectively. At a bass, low pitch (i.e., grave) sound produced by tabla or dholak becomes predominant, while at treble, high pitch (i.e., shrill) sound produced by flute or ghoonghroo (ankle bells) becomes predominant.

Pitch of a note depends on its frequency. Two notes sounded on the same instrument with same amplitude, will differ in pitch when their vibrations are of different frequencies.



(a) Low Pitch Note (Frequency = f)

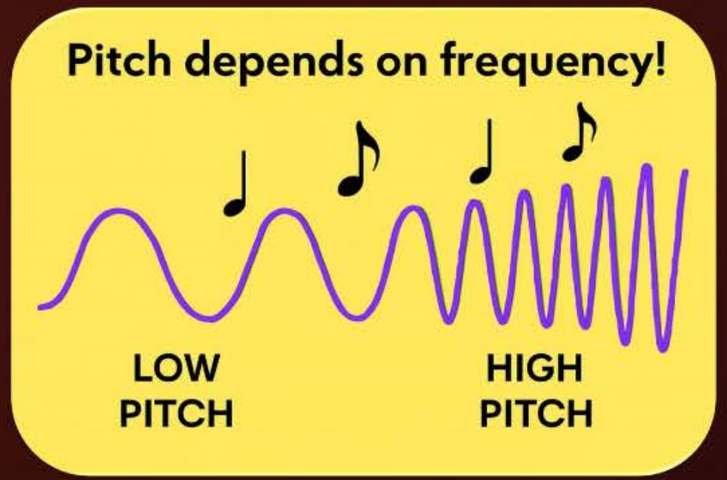


Subjective Nature of Pitch and Objective Nature of Frequency

↳ Depends on observer.

↓ Measurable Quantity.

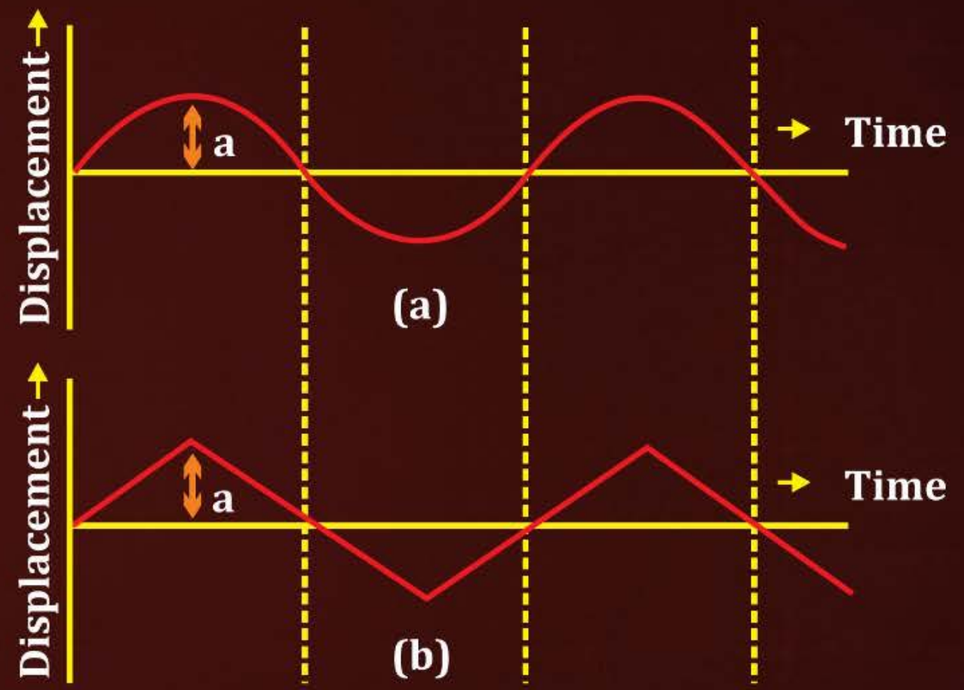
- Pitch is not the same as frequency.
- The pitch refers to the **sensation as perceived by the listener**.
- It may be different for a sound of a particular frequency to the **different listeners** (i.e., pitch is subjective).
- On the other hand, frequency is a measurable quantity. It depends on the source producing the sound.
- It has a definite value for a given sound which is equal to the frequency of its source.
- It has nothing to do with the listener, so frequency is an objective quantity.



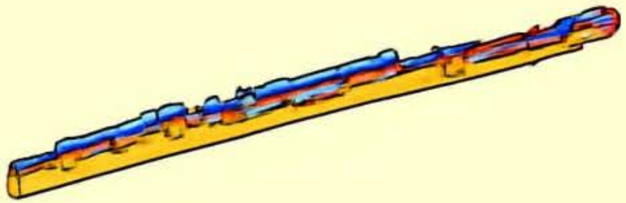
Quality (or Timbre) and Wave Form

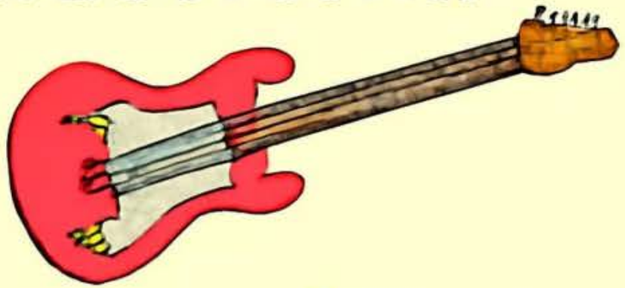
Quality (or timbre) of a sound is that characteristic which distinguishes the two sounds of the same loudness and same pitch, but emitted by two different instruments.

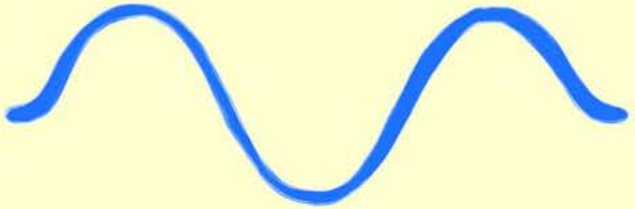
Fig. shows the wave patterns of two sounds of same loudness (i.e., same amplitude) and same pitch (i.e., same frequency), but emitted by two different sources. They produce different sensations in our ears because they differ in wave form, one is a sine wave while other is a triangular wave.

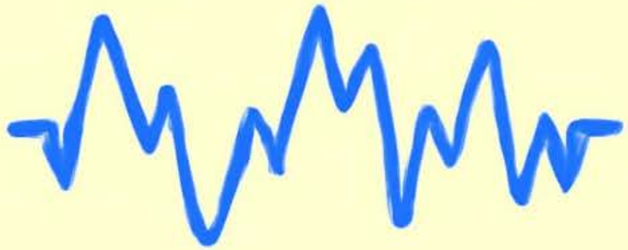


Quality (Timbre)
depends on waveform!









Soft Timbre

Harsh Timbre



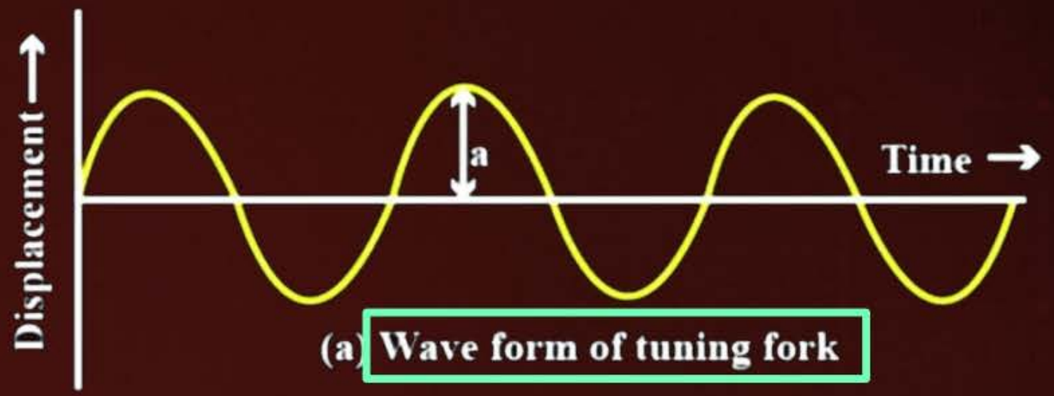
Quality (or Timbre) and Wave Form

The quality of a musical sound depends on the wave form.

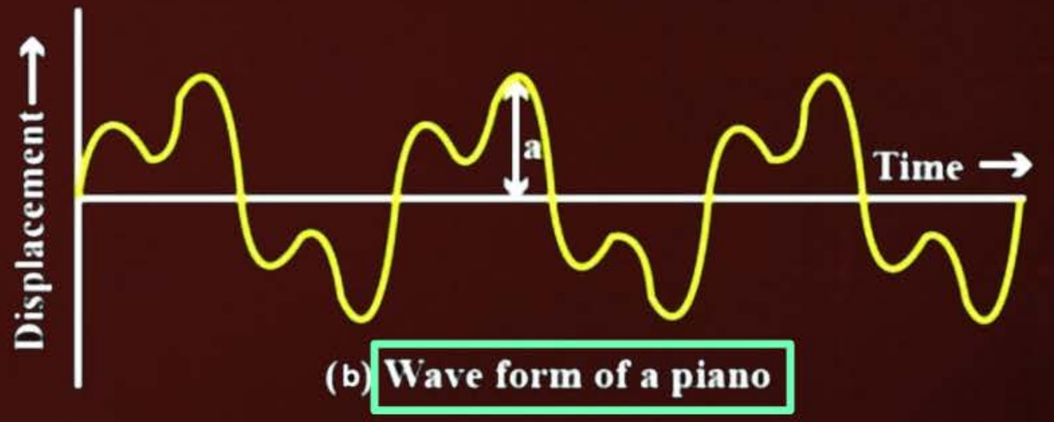
- Usually the sound from an instrument does not contain a note of single frequency, but it contains a combination of vibrations of different frequencies and different amplitudes.
- The vibration of lowest frequency and maximum amplitude is called the **principal (or fundamental) vibration** and the vibrations of frequency integer multiples of it, are called the **subsidiary (or secondary) vibrations**. The subsidiary vibrations are of small amplitudes.
- The resultant vibration obtained by the superposition of all these vibrations, gives the wave form of the sound which we hear.

Quality (or Timbre) and Wave Form

- The sketch (a) shows a pure note (sine curve) of single frequency produced by the tuning fork.



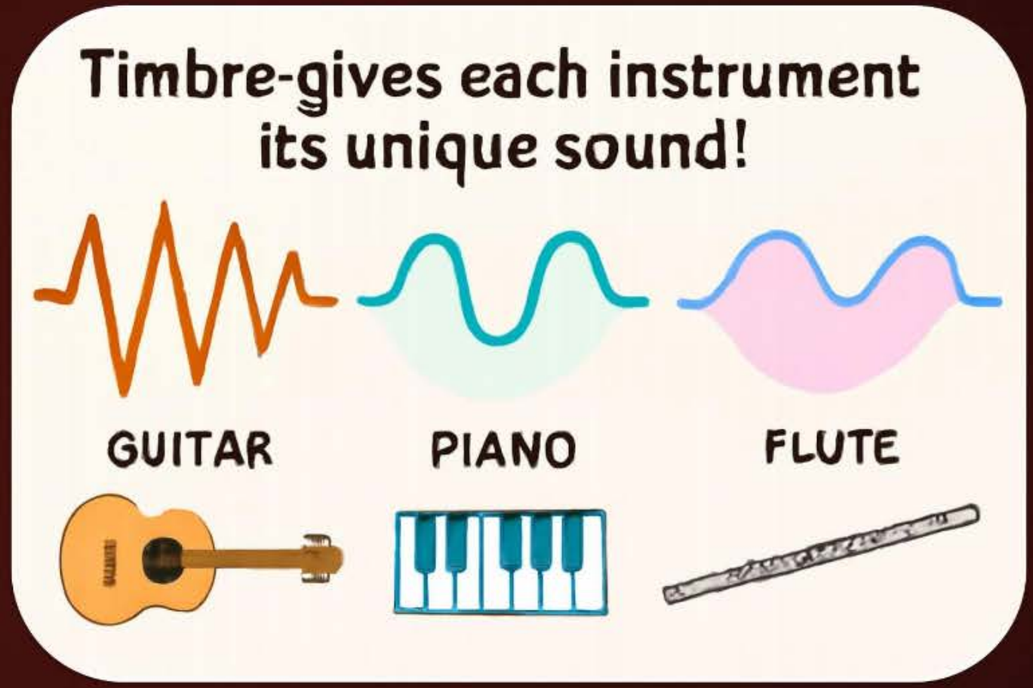
- While the sketch (b) is of same amplitude and same frequency (or same pitch), but it has a different wave form due to the presence of a mixture of subsidiary vibrations along with the principal vibration. The subsidiary vibrations present in the musical note make the wave form complex. It is not a sine curve.



Quality (or Timbre) and Wave Form

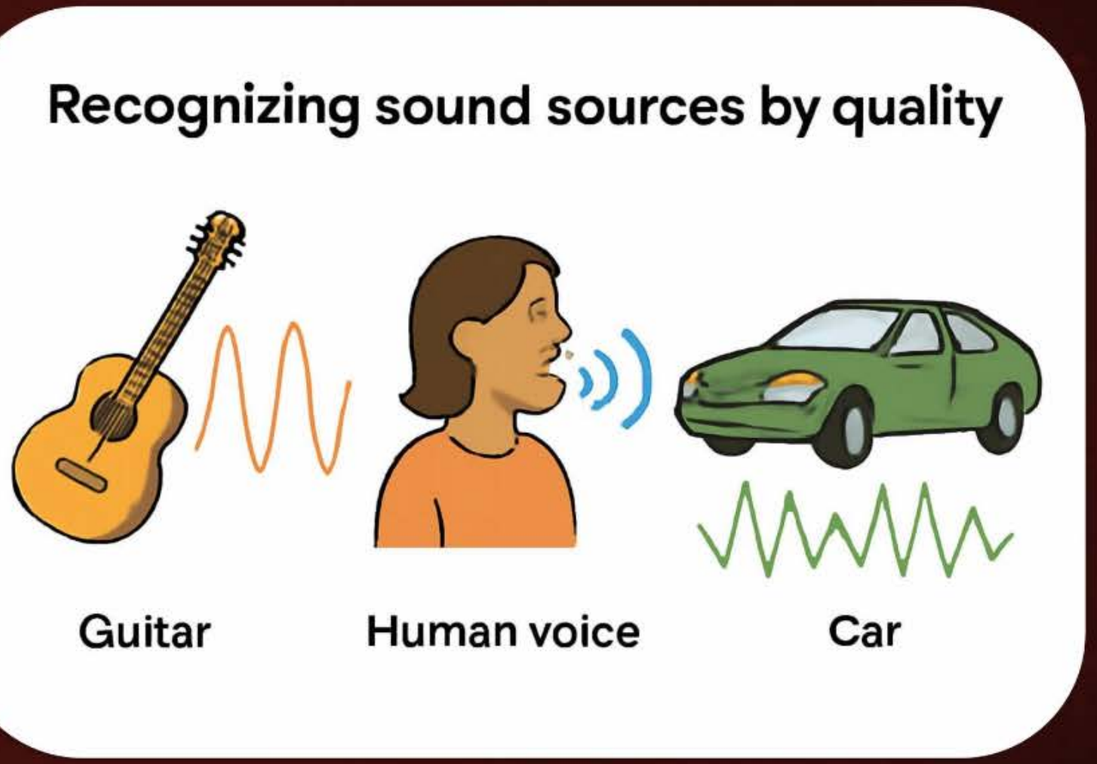
- **Thus, the quality of a musical sound depends on the number of the subsidiary notes and their relative amplitudes present alongwith the principal note.**

- Different instruments produce different subsidiary notes. A note played on a piano has a large number of subsidiary notes, while the same note when played on a flute contains only a few subsidiary notes. Thus, we can easily distinguish between the sounds of a piano and a flute by their different wave forms, though they may be of exactly the same loudness and same pitch.



Recognition Of The Sources Of Two Sounds By Their Quality

- Each vibrating body has its characteristic wave form.
- This makes it possible for one to recognize the vibrating body even without seeing it.
- You generally recognize a person by hearing his voice on telephone without seeing him. It is because the vibrations produced by the vocal chord of each person have a characteristic wave form which is different for different persons.
- Similarly, one can distinguish and recognise the sounds of two different musical instruments because of their different wave forms even if they are of same pitch and same loudness.

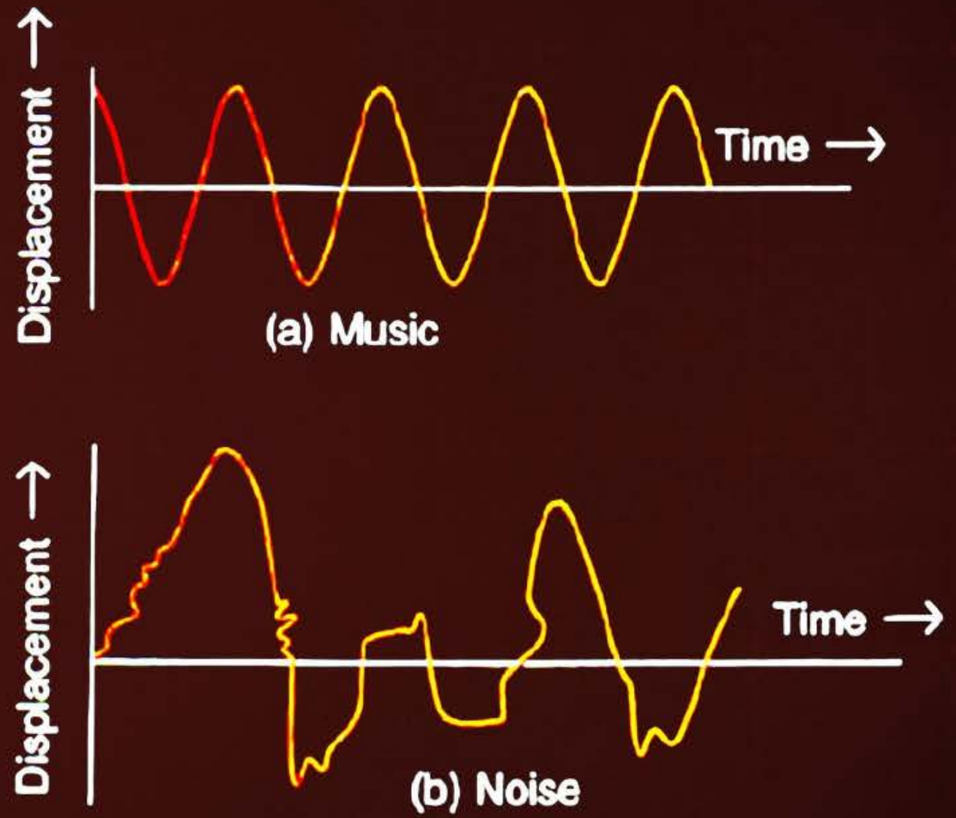


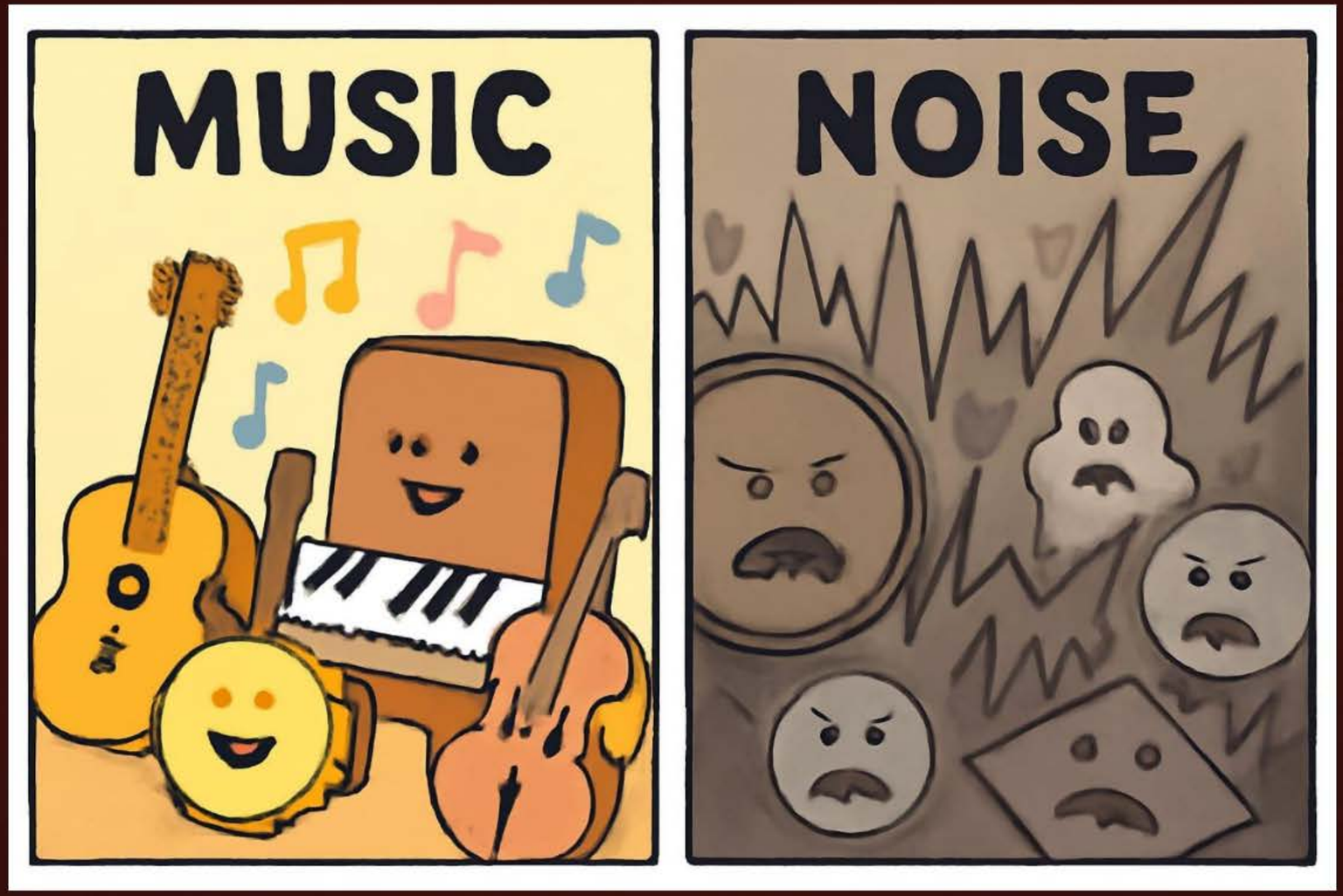
Music And Noise

All sounds, which produce the sensation of hearing, can be roughly divided into two categories: (1) music, and (2) noise.

(1) Music: It is a pleasant, continuous and uniform sound produced by the regular and periodic vibrations. For example, the sounds produced by a violin, piano, flute, tuning fork, etc. are the musical sounds. Their sound level is usually between 10 dB to 30 dB.

(2) Noise: The sounds other than the musical sound are called noise. It is a sound produced by an irregular succession of disturbances and is usually a discontinuous sound. It is discordant and unpleasant to the ears. For example, the sound produced when a stone is thrown on a tin shade is a noise. Usually, all the sounds of level above 120 dB are termed as noise.




MUSIC
NOISE



Comparison Between The Music And Noise



Music	Noise
1. It is regular, smooth and pleasant to the ears. 2. It is produced by the vibrations which are periodic. 3. All the component waves are similar without any sudden change in their wavelength and amplitude. 4. The sound level is low (between 10 dB to 30 dB). 5. The wave form is regular. Example: The sound produced by the musical instruments.	1. It is harsh, discordant and unpleasant to the ear. 2. It is produced by an irregular succession of disturbances. 3. The component waves change their character suddenly and they are of short duration. 4. The sound level is high (above 120 dB). 5. The wave form is irregular. Example: The sound produced by an aeroplane, road roller, industrial machines, etc.

QUESTIONS

A certain sound has a frequency of 256 hertz (f) and a wavelength of 1.3 m. (λ)

(a) Calculate the speed of sound.

(b) What difference would be felt by a listener between the above sound and another sound travelling at the same speed, but of wavelength 2.6 m?

(a) $v = f\lambda$

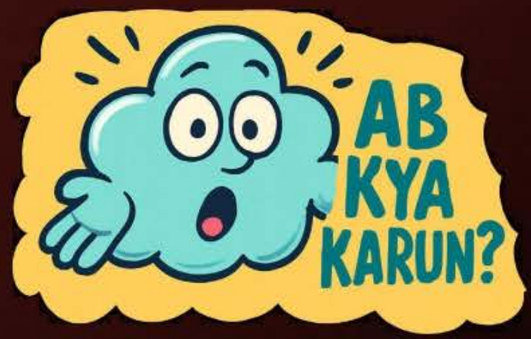
$= 256 \times 1.3$

$v = 332.8$ m/s.

(b) $v = f\lambda$

$332.8 = f(2.6)$

$f = 128$ Hz → Graveness



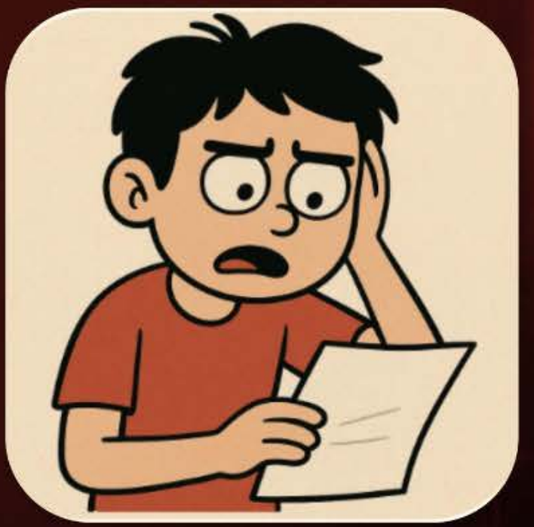
QUESTIONS

The ratio of amplitudes of two waves is 3: 4. Find the ratio of their (i) loudness, and (ii) pitch.

$$\frac{A_1}{A_2} = \frac{3}{4}$$

(i) $\frac{L_1}{L_2} = \left(\frac{A_1}{A_2}\right)^2 = \left(\frac{3}{4}\right)^2 = \frac{9}{16}$ ✓

(ii) $\frac{P_1}{P_2} = \frac{f}{f} = 1:1$



QUESTIONS

By reducing the amplitude of a sound wave, its:

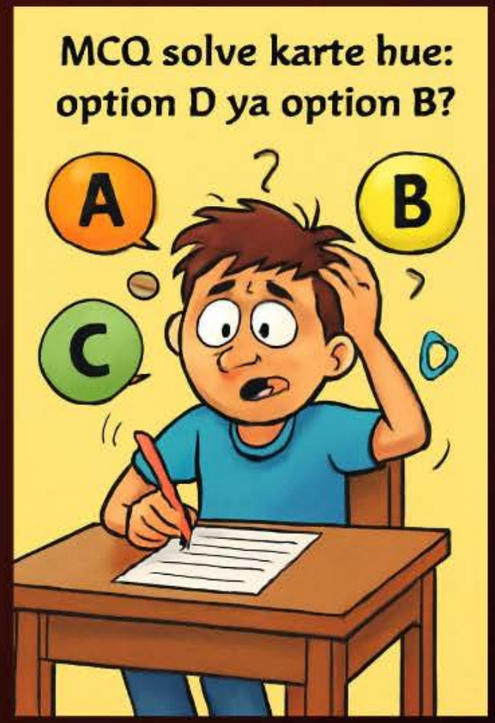
1. pitch increases
2. loudness decreases
3. loudness increases
4. pitch decreases.



QUESTIONS

Two sounds of same loudness and same pitch produced by the two different instruments differ in their:

1. amplitudes
2. frequencies
3. wave forms
4. all the above.

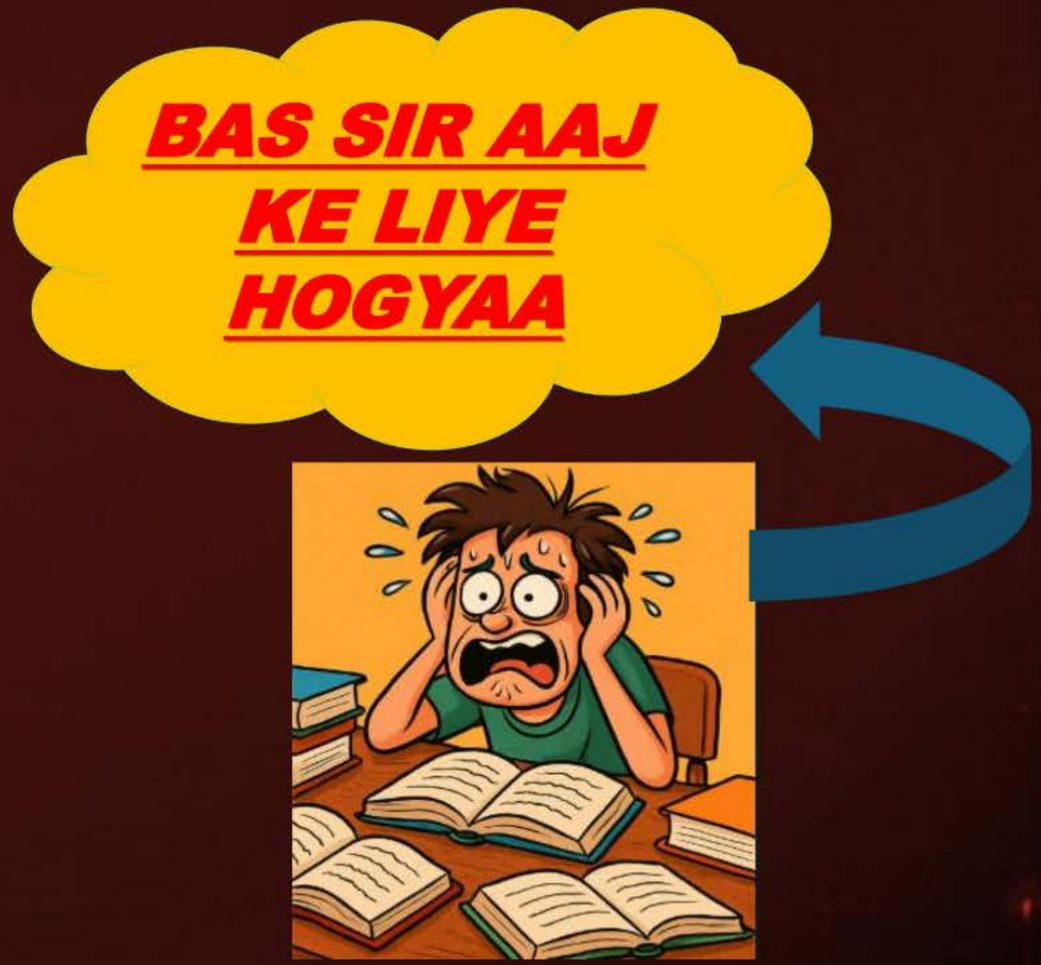


QUESTIONS

Two sounds A and B are of same amplitudes, same wave forms but of frequencies f and 2f respectively. Then:

(A) f (B) 2f

1. B differ in quality from A
2. B is grave, A is shrill
3. B is shrill, A is grave ✓
4. B is louder than A



HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

Thank You